VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Sl.No.	Question

1 In the army recruitment, the height and weights of 10 persons were measured and recorded in the following table.

Height of the person (inches)	5.2	5.6	5.4	5.7	6.1	5.8	5.9	5.2	5.3	5.6
Weight of the same person (kgs)	52	54	48	65	73	65	53	56	58	55

Compute the Karl Pearson correlation coefficient and rank correlation. Why are two coefficients different? Comment on the results.

2. A typing test was conducted for the recruitment of a data entry operator and numbers of mistakes are noted in the following table. Certainly, the test results revealed that the mode and median of typing mistakes distribution are 13.25 and 13.58, respectively. Find the missed frequencies and obtain an expected number of mistakes of the candidates. Also, comment on the distribution concerning the results.

Typing mistakes class	0-3	3-9	9-12	12-15	15-18	18-21	21-24	24-27	27-30
No. of candidates	2	5	32	?	?	?	35	7	4

3. A random variable X has the following distribution

x	-2	-1	0	1	2	3
P(X=x)	k	3k	0.3	2k	0.1	2k

Find (i) the value of k, (ii) $P(X \leq 0)$, (iii) $P(-1.5 < X < 2.5 / X > -1)$ and (iv) mean and variance of the distribution.

4 The amount of kerosene, in thousands of liters, in a tank at the beginning of any day is a random amount Y from which a random amount X is sold during

that day. Assume that the joint density function of these variables is given by

$$f(x,y) = \begin{cases} 2, 0 < x < y, & 0 < y < 1 \\ 0, & elsewhere \end{cases}$$

Find the probability density function for the amount of kerosene left in the tank at the end of the day. Also, test whether random variables X and Y are independent or not?

5. The weights of pigs from two different farms are recorded in the following table

Farm A pig's weights (lbs.)	42	45	52	62	32	52	70	90
Farm B pigs weights (lbs.)	52	68	78	56	68	86		

(i) In which farm, you have observed more growth in weight of pigs?

(ii) In which farm A or B, is there greater variability in individual weight?

(iii) Comment on the Skewness and Kurtosis of both distributions.

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